LAB 3 – COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Pharmacy Expiry & Re-order Dispatch Engine

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Problem Statement	#15 – Healthcare & Telemedicine
Architecture	Layered Architecture

1. UML COMPONENT DIAGRAM

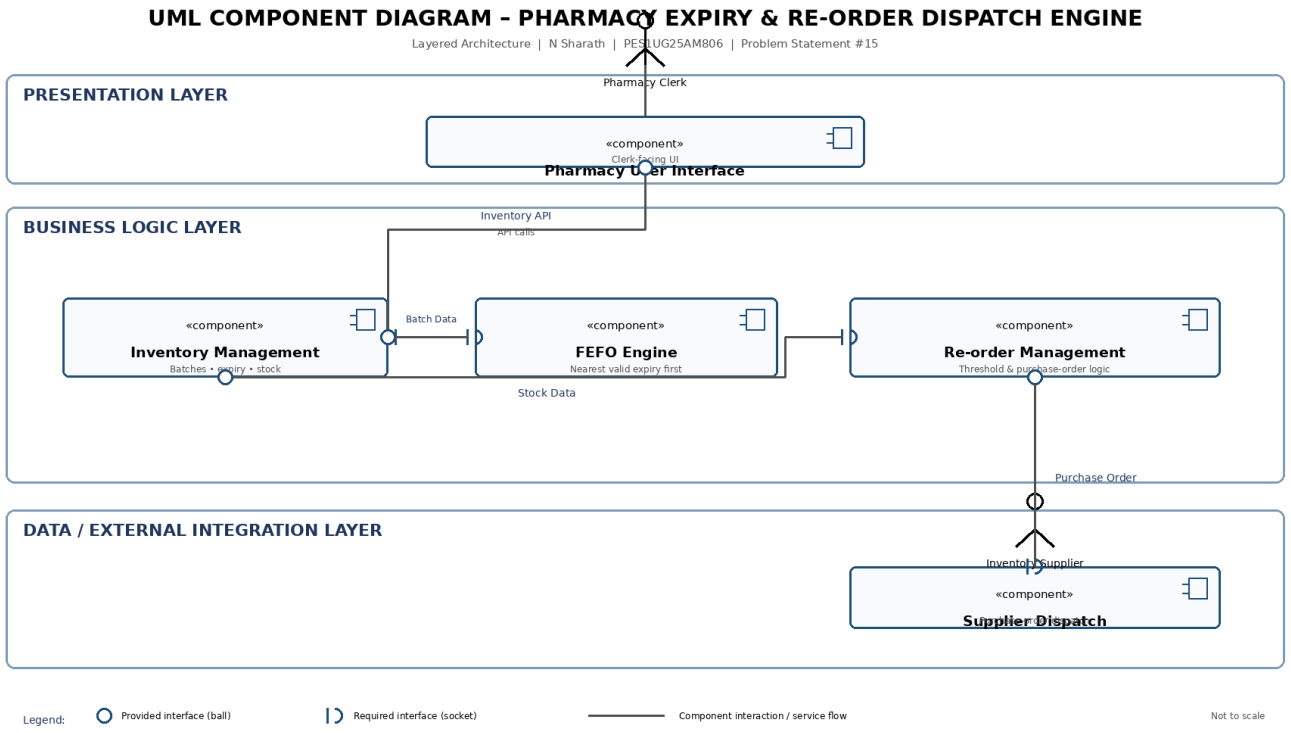


Figure 1. Corrected UML component diagram with five components and four clear component interfaces.

2. ARCHITECTURE SELECTION & WRITTEN JUSTIFICATION

Architecture Selection: We chose **Layered Architecture** for the Pharmacy Expiry & Re-order Dispatch Engine.

Architectural Choice: The system is organized into a Presentation Layer, Business Logic Layer, and Data/External Integration Layer. This provides clear separation of responsibilities and maps naturally to pharmacy operations.

Reason 1 – Separation of concerns: User interaction, inventory/FEFO/re-order business rules, and external supplier operations are kept separate. This makes the system easier to understand, test, maintain, and modify.

Reason 2 – Scenario suitability: Batch tracking, FEFO selection, stock-threshold monitoring, and purchase-order dispatch are well-defined business operations. A layered structure supports these operations without the additional operational complexity of microservices.

Security Advantage: Inventory and supplier operations can be controlled at the business layer. The presentation layer does not directly access stored data; requests pass through controlled business services.

Performance Benefit: Inventory and FEFO operations can be optimized within the business layer and database access can be tuned independently. This avoids unnecessary distributed-service network overhead for this scenario.

Architecture Comparison: Layered Architecture was selected over Microservices and Client-Server. Microservices offer independent scaling and fault isolation but introduce operational complexity, network latency, and data-consistency challenges. Client-Server offers centralized control but has single-point-of-failure and scalability limitations.

Component Mapping: Pharmacy User Interface → Inventory Management → FEFO Engine / Re-order Management → Supplier Dispatch. Interfaces shown are Inventory API, Batch Data, Stock Data, and Purchase Order.

Conclusion: The proposed layered component architecture provides modularity, clear interfaces, controlled data access, and suitable performance for the assigned pharmacy system.

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